Jack's PDF

Product requirements document

IMPLEMENTED BASELINE / VERSION 1.9 / NOVEMBER 5, 2026

A private, minimalist Mac app for reading one PDF at a time, moving easily through a folder, and saving rotated copies while protecting the source.

Product owner / creator	Nikhil Sharma
Implementation	Native Swift, AppKit and PDFKit
Platform	macOS 14 or later; current release binary for Apple silicon
Document status	As-built specification for the current public release
Licence	MIT
Development credit	Created with OpenAI ChatGPT models

Product intent

Make everyday PDF reading calm and predictable: fit the page to the available width, keep controls compact, make the active page explicit, and avoid accidental edits or overwrites. The app is a personal utility being released as open source.

Primary user and jobs

- A Mac user who wants to open and read a PDF quickly, without an editing workspace.
- A user reviewing several PDFs in a folder with the arrow keys.
- A user correcting page orientation and exporting a separate rotated copy.

Scope boundary

The only document modification is page rotation followed by Save As. Printing, sharing, cloud uploads, form filling, annotations, text editing, tabs, multiple document windows and a thumbnail sidebar are outside the product scope.

This document describes the shipped behaviour, not every earlier design proposal. Acceptance evidence and known limits are separated on page 9.

PRODUCT REQUIREMENTS / 02

Opening and reading

One running app instance, one reader window and one active PDF. A new successful open replaces the current document.

ID	Requirement / acceptance behaviour
R01	Open through Finder, the empty reader, File → Open PDF, or File → Open Recent. Reuse the existing app and window. Accept one PDF per open request.
R02	Start explicit opens in Fit mode. Preserve the selected zoom mode when using Left / Right to move between neighbouring PDFs. Never reflow text or change PDF text placement.
R03	Show pages continuously in a vertical scrolling view, with subtle separation. Up / Down scroll toward earlier / later content. No page-preview pane.
R04	Click visible paper to make that page active, even when several pages fit on screen at low zoom. Do not move the viewport or change zoom. Show a thin accent outline; Page x of y, Fit and Rotate follow that page.
R05	Actual scrolling resumes normal page tracking. Clicking empty canvas preserves the chosen page. Explicit focus survives zoom and rotation.
R06	Left / Right open the previous / next PDF in the current folder. Stop at the first and last entries; do not wrap. Use the selected filename or creation date-time order; the default is natural filename order, so 2.pdf precedes 10.pdf.
R07	Count visible, readable, regular .pdf files directly in the selected folder. Ignore case in the extension; exclude hidden files, symbolic links, packages and subfolders.
R08	Request only the containing folder for counting and neighbour navigation. Explain optional access in the native folder picker, showing filename and folder path. Cancel keeps the PDF readable; Browse folder… enables access later.

Read-only interaction

Permit text selection and copying when the PDF allows it. Suppress form editing, annotation editing, embedded document actions, activated links, printing and PDF-content context menus. Existing form values remain visible.

Failed or cancelled opens

Keep the current PDF and any unsaved rotations until another PDF loads successfully. A password prompt, cancelled file picker or failed load must not silently close the current document.

PRODUCT REQUIREMENTS / 03

PDF order and navigation

A subtle dropdown on the existing PDF counter gives users a predictable choice of filename or creation date-time order.

ID	Requirement / acceptance behaviour
F01	Click the borderless PDF x of y counter or its small downward chevron. Offer Filename (A-Z) and Creation date & time (newest first), with a checkmark on the selected choice. Retain 13-point Avenir, fixed neighbouring anchors and a tooltip explaining arrow directions.
F02	Default to natural filename order. Remember one global preference between launches and apply it to all approved folders. An absent or unknown stored preference falls back to filename order.
F03	For creation order, compare the original files' full filesystem creation timestamps, including available fractional seconds. Do not use modification time, PDF authoring metadata, snapshot timestamps or formatted date strings. Timezone changes must not alter the order.
F04	Place newest timestamps first. Left moves to newer PDFs; Right moves to older PDFs. Equal timestamps use natural filename order; exact path ordering resolves equivalent filename comparisons. Missing or invalid timestamps appear last in filename order.
F05	Changing order keeps the open PDF, active page, scroll position, zoom and unsaved rotations. Recalculate x as the file's position in the chosen order; y changes only when the eligible folder contents change. Recent history remains ordered by successful opening time.
F06	Refresh the folder before choosing a neighbour. Match the loaded file by path and filesystem identity; a unique identity match can retain its place after a rename. Do not mistake a replacement at the old path, or ambiguous hard links, for the loaded document.
F07	Stop at both ends without wrapping. With one eligible PDF, arrows leave it open. Failed or cancelled next-file loading preserves the current document and rotations. Ignore stale background scan results after newer scans or access changes.

Unavailable files and permissions

Without folder access, retain Browse folder… and its existing permission flow. If the open PDF is no longer in the list, show PDF - of y (an unavailable-position mark in the app), explain the condition on hover, and request reopening before navigation. Do not silently select another file.

If the folder cannot be read, show Folder unavailable. The dropdown additionally offers Allow Folder Access… to retry the native folder permission flow. Keep the loaded PDF readable. Never silently switch to modification-date order.

Scope and performance

Read folder metadata in the background; do not inspect PDF contents to determine order. Reuse the refreshed list during the next-file load. The preference does not change document bytes, folder permissions or recent-file history.

PRODUCT REQUIREMENTS / 04

Interface and menus

Dark, compact and consistent. All app-owned text uses 13-point Avenir; PDF page typography and native macOS panels keep their own presentation.

Header and window

• Title bar: close and minimize buttons, a small colourful app icon, then Jack's PDF. Hide the green window button.

• Document header: filename; then right-aligned file size, folder position (PDF x of y) and page position (Page x of y), separated by bold bullets. The folder counter opens the PDF-order dropdown. Reserve its small chevron and keep metadata anchors stable during navigation and order changes.

• Below the filename, show the containing folder path only. Long names and paths use the available space and allow horizontal scrolling; tooltips reveal the full text.

• Controls: split Rotate button, Fit, minus, zoom slider, plus; labels read Min. 20%, current percentage and Max. 200%. Fit text stays centred.

• Outside full screen, use a 16:10 window with space around it and a minimum size of 800 × 500 points. Reflow the header at narrower widths without shrinking its text.

Final menu structure

Menu	Available commands
Jack's PDF	About Jack's PDF; Shortcuts; Quit Jack's PDF
File	Open PDF…; Open Recent; Save As…; Close PDF
Open Recent	Up to ten filenames; Clear History. Show No Recent PDFs when empty.
View	Full Screen; changes to Exit Full Screen while active.

There is no Window menu and no Hide menu item. Shortcuts opens a compact reference with no scrollbar. Enable Open, Save As, Close Window, Quit, Full Screen, zoom, Fit, 100% and current/all-page rotation keys. Preserve arrows, Page Up/Down, copy/select-all and dialog keys. No Find or Save shortcut.

Full screen

Use native macOS full screen, temporarily releasing the normal window aspect constraint. Exit restores normal window behaviour. The View menu contains only the relevant full-screen command.

Finder integration

Register as a PDF viewer and supply the colourful PDF document icon. Choosing the default PDF app remains a user-controlled macOS action. Finder page thumbnails and custom file icons can take precedence; the app does not change Finder preview preferences.

PRODUCT REQUIREMENTS / 05

Zoom, rotation and export

Preserve document structure and protect the original. Rotation is temporary viewing state until a separate copy is exported.

Area	Required behaviour
Fit and manual zoom	Fit the active page to the viewport width. Recalculate on resize and relevant rotation, without changing Fit merely because scrolling selects another page. Manual zoom spans 20-200% in five-percentage-point steps; Fit may exceed those limits.
Rotation control	The main button rotates the active page 90° clockwise. Its dropdown offers Rotate current page and Rotate all pages. Support repeated clicks and full turns.
Animation	Use a smooth 600 ms clockwise transition with subtle perspective and shadow movement. Continue clockwise through complete turns; do not reverse at 270° to 0°. Queue repeated actions and respect Reduce Motion.
Save As	Offer the native save dialog after rotations. Start in the source folder with a suggested new filename. There is no Save command. Reject the original, including equivalent paths and hard links.
Other destinations	Allow replacing another eligible regular file after the normal macOS confirmation. Validate destination, access and source identity; handle failed writes without damaging the source.
Quality	Apply native PDF page orientation. Do not rasterize pages, downsample embedded images or convert them to JPEG during export. Preserve native text/vector content.

Unsaved rotation decisions

Quit or close the app window: ask whether to Save As, Discard or Cancel. Cancelling keeps the document and rotations open.

Switch PDFs or choose Close PDF: do not show that warning. Successful replacement or Close PDF discards temporary rotations; a failed or cancelled replacement preserves them.

Document limitations

Open encrypted PDFs using a password prompt; do not store the password in history. Rotated export of encrypted PDFs is unavailable because preserving protection has not been implemented. PDFKit may reorganize PDF structure or metadata on export. Signature validity and specialised PDF conformance are not guaranteed.

PRODUCT REQUIREMENTS / 06

Recent files and permissions

Keep history useful without requiring broad filesystem access. Remembering a file and retaining permission to reopen it are separate responsibilities.

History contract

• Retain the last ten successfully opened PDFs, newest first and deduplicated by normalized file URL. Reopening moves the file to the top.

• Save the filename/path even if macOS cannot create a persistent permission bookmark. Never silently omit a successfully viewed file for that reason.

• Use read-only, app-scoped bookmarks where available. Migrate old preferences only after encrypted storage is written and verified.

• Resolve saved bookmarks to track moved files where possible, then try the remembered path. Renew the bookmark after successful opening. Do not reorder history on failed attempts.

Condition	User experience / state rule
Saved access works	Open directly with no additional permission prompt.
File missing or access lost	Show Locate File…, Remove from History and Cancel. Keep the current document and rotations intact.
Locate File…	Use the native single-PDF picker to renew access or choose the file at its new location. Replace the existing history entry only after a successful load; deduplicate the destination.
Remove from History	Remove that saved reference only. Do not delete the PDF or close a currently displayed document.
Cancel or failed locating	Preserve the history entry and current document.
Clear History	Require explicit confirmation that recent history and all saved folder approvals will be removed. Default to Cancel. Confirming ends folder navigation; the open PDF and rotations remain. Do not delete PDFs.

Folder access stays separate

Opening a recent PDF must not force a new folder grant. Reuse an existing valid folder bookmark if available; otherwise show Browse folder… for optional counting and neighbour navigation. A file permission does not imply access to its entire folder.

Retain ten recent PDFs and up to 10,000 folder approvals, evicting the least recently used approval at capacity. Records have no time-based expiry. Activate only the current folder. Cancelling folder access must not prompt again on each arrow press.

PRODUCT REQUIREMENTS / 07

Privacy, safety and performance

Use native macOS capabilities and narrowly scoped access. The product has no account, tracking service, cloud dependency or network entitlement.

Privacy and original-file protection

• Enable App Sandbox and the hardened runtime. Use standard Open/Save pickers and app-scoped read-only bookmarks. Do not request Full Disk Access, Contacts, Photos, camera or microphone permissions.

• Encrypt recent references and up to 10,000 folder approvals using AES-256-GCM in private Application Support storage. Keep the random key in the local login Keychain and PDF passwords only in memory.

• Read from a private file-backed snapshot so rendering and export stay consistent if the source changes. Use owner-only permissions for temporary directories and PDF files.

• Never export to the original file, including alternate paths, symbolic-link aliases and hard links. Recheck source/destination identities and validate the finished export before placement.

• Remove released snapshots and failed export temporaries. Clean abandoned snapshots on a subsequent load while preserving active ones. This is ordinary cleanup, not secure erasure.

Read the encryption key at startup. Changed ad-hoc signatures, one-time approval or a locked Keychain may require macOS approval. Preserve unreadable storage; never fall back to plaintext. Clear History forgets saved references without deleting the key. The non-sensitive PDF-order preference is stored separately in local app preferences; it contains no paths or timestamps and survives history clearing.

Implementation and performance requirements

Component	Approach / intent
Rendering and UI	Native Swift, AppKit, PDFKit and Core Animation. No Rust or third-party runtime is required. PDFKit renders file-backed pages on demand.
Large-file loading	Prefer APFS copy-on-write snapshots. On unsupported filesystems, copy in the background with a 1 MiB buffer instead of loading the entire PDF into RAM.
Responsiveness	Keep long file preparation away from the main UI thread. Keep rotation snapshots bounded and release temporary animation resources.
Quality and scope	Optimise for interactive reading and smooth scrolling. Do not claim a universal memory, opening-time or frame-rate target without representative benchmarks.

Public-release privacy

Publish source, generated-test code, documentation and intended artwork only. Exclude private PDFs, local history/bookmarks, machine paths, credentials and signing identities. Use GitHub's no-reply commit address. Strip camera/editing metadata from the public Jack photo while preserving its decoded pixels.

PRODUCT REQUIREMENTS / 08

About and key messages

Maintain the same compact typography and clear recovery choices throughout the app. Native system dialogs retain macOS wording and presentation.

About content order

Centred app icon → text divider → Jack's PDF → divider → short description → divider → creator/model/date/version credits → divider → memorial → photograph.

Use a nonresizable panel with no scrolling. Compact vertical gaps and a 48-point icon keep all contents visible. The original photograph appears proportionally in a small, centred rounded frame; its display is 128 × 96 points.

Content	Required text
Description	A private, minimal PDF viewer for Mac.
Credits	Created by Nikhil Sharma Made with OpenAI ChatGPT models. Last updated: November 5, 2026 Version 1.9
Memorial	In loving memory of Jack, our sweetest companion and forever friend. You are still in our hearts and our lives every day.

Shortcuts reference

Place Shortcuts below About and above Quit. List every supported reader and dialog shortcut in an aligned two-column, nonresizable modal. Pair related commands where useful; retain 13-point Avenir and show all content without scrolling. Include the modifier-key legend and folder-access/copying limitations.

Important messages and decisions

Original-file protection: "To prevent accidental overwriting, original PDF is always protected. Choose a different filename."

Recent-file recovery: "Locate this recent PDF?" Explain that the file may have moved, be unavailable, or need permission again. Offer Locate File…, Remove from History and Cancel.

Closing with rotations: "Keep your page rotations?" Explain that the original has not changed and a rotated copy can be saved before closing. Offer Cancel, Discard and Save As….

Clear History: explain that recently opened files and saved folder permissions are forgotten. Offer Cancel as default and Clear History and Folder Access as the explicit destructive choice.

Errors: explain unreadable/empty PDFs, password failures, inaccessible files, invalid destinations and failed saves without losing the current document. Disable document controls when no PDF is open or work is in progress.

Folder-order labels and recovery states are specified on page 3. The accompanying app-text review is the detailed wording inventory. Dynamic filenames, counts and native macOS/provider messages are not fixed product copy.

PRODUCT REQUIREMENTS / 09

Acceptance and release

Version 1.9 is the implemented baseline. This PRD records tested behaviour and explicit limits; it does not convert spot checks into performance guarantees.

Acceptance area	Evidence and boundary
Opening / navigation	Checks cover single-instance reading, natural and creation-time folder order, fractional/tied/missing timestamps, rename/replacement identity, boundaries and saved preferences. Live checks verified both arrow directions and the actual dropdown.
Zoom / focus / rotation	Native reader checks covered Fit/manual zoom, active-page selection, repeated clockwise turns, rapid-click queuing, layout stability and blocked editing/printing paths.
Original and output fidelity	DocumentSupport checks passed original/path/link protection, native text and existing form content, unchanged decoded scanned-image pixels, encrypted-export refusal and snapshot cleanup.
History / storage (1.8)	Checks passed migration, authenticated encryption, tampering/wrong-key rejection, file modes, fresh nonces, 10,000-folder round trips, LRU eviction and durable clearing. Live checks verified approval, update reauthorization and explicit clearing.
Interface / state	Version 1.9 checks passed 68 layouts at 800-1186 points with virtual counts through 1,000,000 PDFs. Order changes preserve document, page, scroll, zoom, rotations and history. Earlier full-screen and non-scrolling modal checks remain recorded.
Validation limits	No universal performance benchmark, full accessibility certification, notarization, digital-signature preservation claim or comprehensive test of every PDF producer. macOS 14 is the deployment target, not a claim of tests on every supported OS.

Public release

Repository: github.com/nikhilrsharma/jacks-pdf
Release baseline: 1.9 · Creator: Nikhil Sharma · Licence: MIT

Provide readable source, build/test instructions, privacy notes, this PRD and a versioned release. The supplied binary is an Apple-silicon build with local ad-hoc signing, not a Developer ID-notarized distribution. Users can build from source with Xcode; an Intel build is not validated by the current binary checks.

Source of truth and future changes

Baseline reviewed against Sources/App.swift, ReaderPDFView.swift, ZoomControl.swift, ReaderDialog.swift, DocumentSupport.swift, EncryptedLibrary.swift, ReaderShortcut.swift, ShortcutReference.swift, Info.plist, Viewer.entitlements, README.md, SECURITY.md, test sources and VALIDATION.md. All are repository-relative references; no private user documents were used in this PRD.

Future work requires a separate product decision. Search, printing, sharing, editing, tabs, automatic updates and cloud sync are not implied by this release.